

Netflix Movie Database Analysis

An end-to-end data analysis of a comprehensive movie database containing 9,827 entries, exploring genre distribution, popularity outliers, and rating patterns.

Project Overview & Dataset Characteristics

Dataset Foundation

This project involved comprehensive analysis of the `mymoviedb.csv` file, a robust dataset containing detailed information about movies across multiple dimensions. The initial inspection revealed a dataset with significant depth but requiring substantial cleaning and transformation.

The dataset contained **9,827 total rows** representing individual movie entries, with **9 columns** of information including `Release_Date`, `Title`, `Overview`, `Popularity`, `Vote_Count`, `Vote_Average`, `Original_Language`, `Genre`, and `Poster_Url`.

Initial Data Quality Assessment

The dataset presented with considerable complexity. While rich in information, it was initially very messy with many missing values that required attention. The inspection revealed **100 missing values** specifically in the `Vote_Average` column, which were subsequently dropped to ensure statistical accuracy for analysis.

This data quality issue represented approximately 1% of the total dataset, a manageable percentage that could be addressed through careful data cleaning without significantly impacting the overall analysis scope.

Data Cleaning & Transformation Process

Several critical transformation steps were executed to prepare the raw data for meaningful visualization and analysis:

01

Temporal Conversion

The Release_Date field was converted to `datetime64[us]` format, enabling chronological analysis. The year component was then extracted as an integer to facilitate time-based trends and historical comparisons.

03

Genre Normalization

The Genre column contained comma-separated strings (e.g., "Action, Adventure"). These were split into lists and "exploded," resulting in a new expanded dataset of **25,552 rows**, where every movie-genre pair occupies its own row.

02

Dimensionality Reduction

Irrelevant columns for this specific analysis—Overview, Original_Language, and Poster_Url—were systematically dropped to streamline the dataframe and focus on core analytical dimensions.

04

Categorization

Using a custom `categorize_col()` function, the Vote_Average was binned into four distinct categories based on quartiles: `not_popular`, `below_average`, `average`, and `popular`.

Statistical Insights: Rating & Popularity Metrics

6.44

Average Rating

Mean Vote_Average across the entire dataset

40.33

Mean Popularity

Average popularity score with massive variance

108.87

Popularity Std Dev

Indicates significant outliers in data

Popularity Distribution Analysis

The mean popularity score of 40.33 is dramatically influenced by extreme outliers, as evidenced by the massive standard deviation of 108.87. This indicates that while most movies cluster around lower popularity scores, a small number of blockbusters achieve extraordinarily high popularity ratings.

The standard deviation being nearly 3 times the mean suggests a highly skewed distribution, where typical movies have modest popularity while a select few achieve viral-level attention. This pattern is consistent with the "power law" distribution commonly observed in entertainment industries.

Vote Distribution Balance

The categorization process resulted in a balanced distribution across the four rating tiers, with "not_popular" holding 2,467 entries and "popular" holding 2,450 entries.

This balanced quartile-based categorization ensures that each popularity tier contains approximately 25% of the dataset, enabling meaningful comparative analysis across popularity levels.

Extreme Outlier: Maximum Popularity

Top Movie Spider-Man: No Way Home (2021)	Popularity Score 5,083.95	Context Massive outlier representing extreme statistical deviation
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The most popular movie in the dataset, **Spider-Man: No Way Home** released in 2021, achieved a staggering popularity score of **5,083.95**. This represents an extraordinary outlier that is more than 100 times the mean popularity score of 40.33.

This extreme value illustrates the massive disparity between typical films and global blockbusters. The popularity score for Spider-Man: No Way Home is so high that it significantly skews the overall distribution, which is why the standard deviation reaches 108.87 despite the relatively modest mean.

Such outliers are critical to identify because they can distort statistical analyses if not properly accounted for. In this case, the extreme popularity of this single title highlights the "winner-take-all" dynamics of the modern entertainment industry, where a small number of productions capture disproportionate audience attention.

Genre Analysis: Dominance Patterns

Genre Distribution Overview

Analysis of the 19 unique genres revealed significant variation in frequency distribution. The genre normalization process expanded the dataset from 9,827 original movies to 25,552 movie-genre pairs, enabling granular analysis of genre preferences.

This expansion revealed that certain genres dominate the database while others appear relatively infrequently. The most frequent genres represent the backbone of mainstream content production, while niche genres maintain smaller but dedicated followings.

Key Findings

- **Drama** appears 3,715 times, representing the most frequent genre
- Comedy and Action follow as highly frequent genres
- 19 unique genres identified across the dataset
- Multi-genre tagging enables cross-category analysis

3,715

Drama Appearances

Most frequent genre in database

19

Unique Genres

Total distinct genre categories

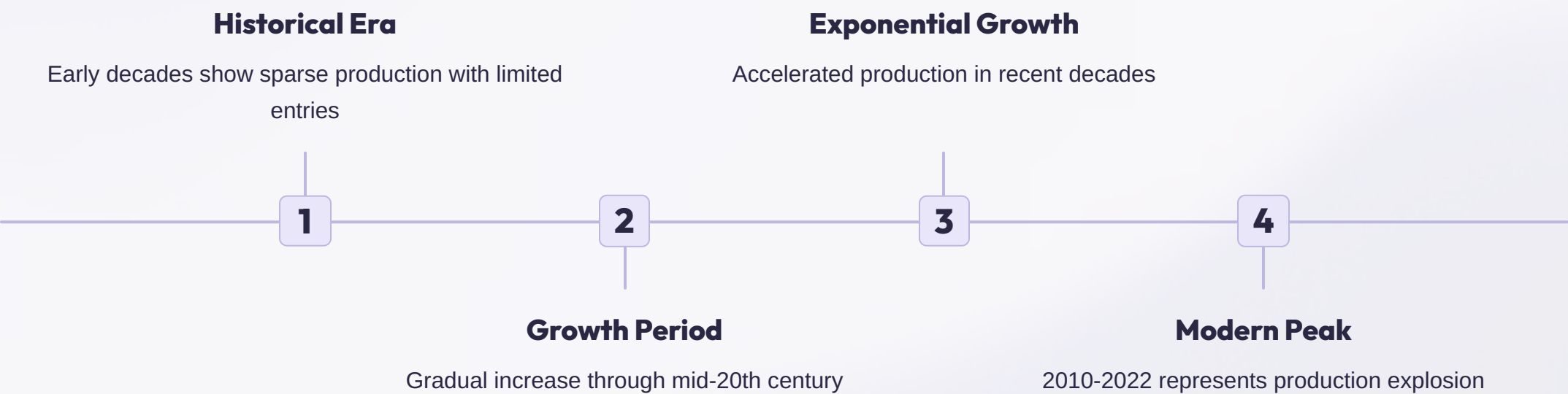
25,552

Movie-Genre Pairs

After normalization process

Historical Production Trends

A histogram analysis of the Release_Date field revealed striking patterns in movie production over time:



Exponential Increase

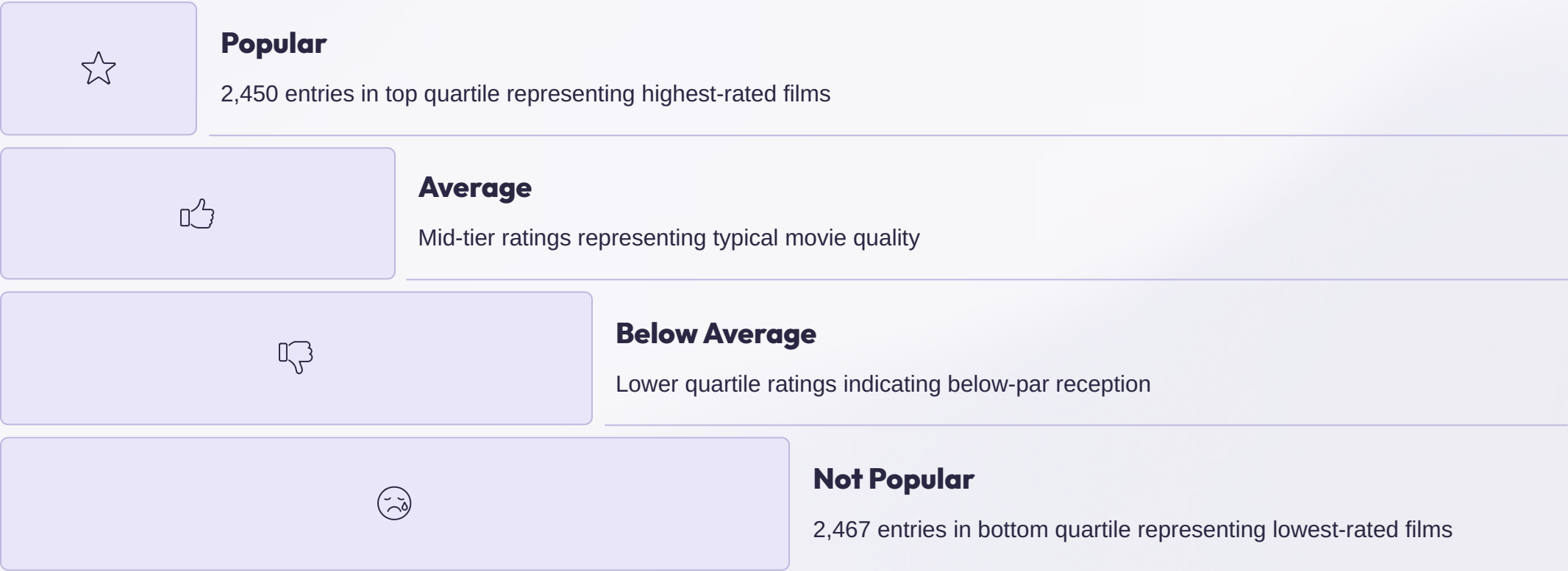
Movie production has seen an exponential increase over the decades, with the vast majority of the 9,827 movies being released between **2010 and 2022**. This concentration in recent years reflects both the digital revolution in film production and the expanding scope of the dataset to include contemporary releases.

Database Bias

The temporal distribution reveals a modern-leaning database, with relatively fewer entries from earlier decades. This bias toward recent productions means that historical trends may be underrepresented in the analysis, and findings should be interpreted with this temporal limitation in mind.

Vote Average Categorization

The Vote_Average was systematically categorized into four distinct tiers using quartile-based binning:



This balanced categorization ensures that each popularity tier contains approximately 25% of the dataset, enabling meaningful comparative analysis across popularity levels. The distribution shows that "not_popular" holds 2,467 entries while "popular" holds 2,450 entries, creating nearly equal group sizes for statistical comparison.

Key Findings Summary



Dataset Scale

9,827 movies expanded to 25,552 movie-genre pairs through normalization



Statistical Insights

Mean rating of 6.44 with dramatic popularity outliers



Genre Dominance

Drama leads with 3,715 appearances across database



Temporal Bias

Concentration in 2010-2022 reflects modern database focus

Technical Achievements

- Successfully cleaned 100 missing values from Vote_Average column
- Implemented custom categorization function for quartile-based analysis
- Expanded dataset through genre normalization for granular insights
- Converted temporal data for chronological trend analysis

Business Insights

- Drama represents dominant content category in modern production
- Extreme popularity outliers like Spider-Man demonstrate winner-take-all dynamics
- Recent production explosion reflects digital revolution in film
- Balanced rating distribution enables comparative analysis

Conclusion & Implications

The analysis successfully transformed a raw CSV file into a clean, categorical dataset ready for visualization and insight generation. The findings reveal a modern-leaning database where **Drama** reigns supreme as the dominant genre, appearing 3,715 times across the 25,552 movie-genre pairs.

High-budget blockbusters like **Spider-Man: No Way Home** represent extreme statistical outliers, achieving a popularity score of 5,083.95 that is more than 100 times the mean popularity of 40.33. This dramatic variance, reflected in the massive standard deviation of 108.87, illustrates the "winner-take-all" dynamics characteristic of the modern entertainment industry.

Methodological Success

The data cleaning pipeline effectively addressed missing values, normalized genre structures, and created meaningful categorical variables for analysis.

Temporal Considerations

The concentration of entries between 2010-2022 suggests the database reflects contemporary production trends rather than historical patterns.

Future Research

Additional analysis could explore correlations between genre, release year, and popularity to identify emerging content trends.

This analysis provides a foundation for understanding content patterns in the Netflix movie database, with implications for content strategy, audience targeting, and production decisions in the streaming era.